- Linus Torvalds, a Finnish computer science student, started Linux as a hobby project in 1991.
- Linux now powers the world's top supercomputers, cloud infrastructure, and even forms the foundation for Android, the leading mobile OS.
- In the server space, Linux has become dominant, powering an estimated 96% of the world's top web servers.

Since its inception as a one-man project in the early 1990s, the Linux operating system has undergone significant expansion. From powering web servers to running supercomputers, its impact is undeniable. But how did this open-source phenomenon come to be? Let's dive into the key milestones in the journey of Linux evolution from a student project to a dominant operating system.

## The Birth of Linux (1991-1994)

The story of Linux began with the concept of Unix, a powerful and versatile operating system. However, Unix was expensive and proprietary, limiting its accessibility. In 1984, a simpler Unix-like system called Minix emerged, sparking the interest of a young Finnish student named Linus Torvalds.

**August 25th, 1991:** Linus Torvalds made a now-famous post to the MINIX newsgroup announcing his work on a free operating system for **386(486) AT clones**. He stated it was "just a hobby, won't be big and professional like gnu." Little did he know, this "hobby" would revolutionize the tech world.

**September 17th, 1991:** Linux version 0.01 was released, containing the initial code and basic functionality. The first usable version of Linux, version 0.02, came out in October 1991, consisting of approximately 10,239 lines of code.

## 1992: The Year Linux Became Open Source

The early development of the Linux kernel saw its distribution restricted by non-commercial licenses. However, a crucial decision was made to the Linux kernel in 1992 with the release of version 0.12. It was relicensed under the GNU General Public License (GPL), marking a significant shift in Linux evolution.

This switch from a non-commercial license to the GPL made Linux open source, allowing anyone to freely use, modify, and distribute the kernel. Linus Torvalds himself considers this decision to be the best he ever made.

## The Rise of Linux

The fledgling Linux kernel found a perfect partner in the GNU Project: a group developing a free and open-source operating system. The GNU Project provided essential components like utilities and Linux tools that, when combined with the Linux kernel, formed the foundation for a complete GNU/Linux operating system.

As Linux grew, the need for complete operating systems built around it also grew. This critical need led to the creation of Linux distributions, or “distros.”

1992 saw the release of MCC Interim Linux, one of the first distributions. Others soon followed:

- Slackware (1993)
- Debian (1993)
- Red Hat Linux (1994)
- SUSE Linux (1994)

These distributions packaged the Linux kernel with other free and open-source software, creating complete operating systems and propelling the Linux evolution forward.

### 1994: Linux 1.0.0 Arrived

On March 14th, Linux 1.0.0 was released, consisting of 176,250 lines of code and providing support for single-processor Intel 80386 (i386) systems.

## Tux: Symbol of Linux

In 1996, Linus Torvalds announced that Linux would have a penguin mascot, inspired by an incident where he was bitten by a little penguin during a visit to the National Zoo & Aquarium in Canberra, Australia. Larry Ewing created the original draft based on this idea. The name Tux, suggested by James Hughes, is a play on “Torvalds’ UniX” and “tuxedo,” reflecting the penguin’s appearance. Since its creation, Tux has remained the symbol of Linux for over two decades.

## 1996: Linux Kernel 2.0

The Linux kernel 2.0 was released, introducing support for multiple processors in a single system (SMP) and more extensive hardware compatibility.

## 1999: Windows Refund Day

On February 15, 1999, in a playful protest known as “Windows Refund Day,” Linux users around the world visited Microsoft offices, attempting to return the unused Windows licenses that came bundled with their computers.

## 1998-2000: Entering Enterprise

The late 1990s and early 2000s witnessed a paradigm shift in the Linux evolution: a surge of interest from the business world. Major tech companies began to recognize the potential of Linux, leading to several key developments:

- 1999: IBM joined forces with Red Hat, announcing support for Linux
- 1999: Dell began pre-installing Linux on select servers
- 2000: The GNOME and KDE desktop environments evolved, making Linux more user friendly for desktop users.
- 2000: IBM invested $1 billion in Linux development.

## 2003: The SCO Group Lawsuit

The SCO Group (SCO) launched a legal assault in 2003, suing IBM for $1 billion. They claimed that Linux contained copyrighted Unix code. This battle, which ultimately failed, underscored the importance of understanding the legal and historical context of Linux's intellectual property.

## 2004: Ubuntu 4.10 Released

On October 20, 2004, Canonical Ltd. released Ubuntu 4.10, codenamed "Warty Warthog," a distribution that made Linux more accessible to non-geek users and significantly boosted its adoption on desktops and laptops.

## 2005: Linus Torvalds' Git

The creation of Git by Linus Torvalds revolutionized version control, becoming an essential tool for software development and marking a significant milestone in the Linux evolution.

## Modern Era and the Future (2011-Present)

Linux has expanded its reach beyond traditional computers, powering mobile devices like smartphones through Android (a mobile OS based on the Linux kernel) and embedded systems that control everything from smart TVs to industrial robots. In the realm of supercomputing, Linux's efficiency and scalability have made it the dominant force. It powers all of the top 500 supercomputers in the world, facilitating groundbreaking research and simulations in various fields.

The rise of cloud computing has further solidified Linux's position. Cloud platforms like Amazon Web Service (AWS) and Google Cloud Platform (GCP) run primarily on Linux for their infrastructure. Additionally, containerization technologies like Docker and Kubernetes, which enable efficient application deployment, are built upon the foundation of Linux.

In April 2015, version 4.0 of the Linux kernel was released, which included significant improvements to live kernel patching capabilities and also sparked discussions about Linux's C library choices, such as musl vs glibc. Live patching allows applying kernel updates without requiring a system reboot. However, it is not an entirely new concept to the Linux ecosystem. Solutions like Oracle's Ksplice have been around for some time, offering live patching exclusively for Oracle Linux users with a Premier Support subscription.

As we look to the future, the Linux evolution continues to thrive, driven by several key trends:

- Increased focus on security and containerization
- Advancements in real-time computing for critical systems
- Continued expansion in emerging fields like AI and machine learning
- Growing adoption in desktop environments

## Final Thoughts

The Linux evolution, from beginning as a student's side project to becoming a cornerstone of modern computing, is a testament to the power of open-source collaboration and innovation. As technology continues to advance, Linux stands at the forefront, ready to adapt and grow. Its adaptability, reliability, and community-driven development have cemented its place in the tech world, promising a future filled with exciting possibilities.

Simplify Linux kernel patching and minimize downtime with **TuxCare's KernelCare Enterprise**. Automate live patching for continuous security. Learn more!

Lear more about Linux on our YouTube series Ep6: https://www.youtube.com/watch?v=wTXAejhM7JA